

Sistema APU — Frontend

Documentation review pack: scope, architecture, traceability, and flagged risks

Repository: `thesis-front-react`

Spec ref: `thesis-docs @ d7508eb`

Plans: 001–015 (all 44 processes / 44 screens)

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TL;DR. The plan set covers the full v1.1 scope — **P-01...P-44** and **S-01...S-44**, each assigned exactly once — across **15 self-contained plans**. Three architectural decisions govern the whole codebase (no client calc engine, decimal-string money, a single API seam). Every plan traces to spec identifiers (P/S/TC/RNF/ADR) so the SPA remains auditable against the thesis. Six gaps were surfaced rather than papered over; a further eight paths were considered and rejected with recorded reasons.

1 · Scope & coverage

The frontend is a React 19 + TypeScript SPA (Vite 8), styled with shadcn/ui 4 on Tailwind 4, using TanStack Query for server state and Zustand for client state, with Vitest + RTL + MSW for unit tests and Playwright + axe-core for E2E and accessibility. Fifteen plans partition the work with hard ordering rules so any plan can be executed from cold context.

#	Plan	Processes	Screens	XP iter.	Status
001	Scaffold + verification baseline	—	—	I-01	DONE
002	API seam: types, client, query layer	—	—	I-01	DONE
003	Testing harness (Vitest · RTL · MSW · Playwright)	—	—	I-01	DONE
004	Design system + decimal formatting	—	—	I-01	DONE
005	Auth module	P-01...P-04	S-01...S-06	I-01, I-02	DONE
006	Shell, routing, global states	P-43, P-44	S-43, S-44	I-01	DONE
007	Proyectos	P-05...P-12	S-07...S-13	I-03, I-06	DONE
008	Insumos	P-13...P-18	S-14...S-19	I-04	DONE
009	APU editor (núcleo)	P-19...P-22	S-20...S-23	I-05	DONE
010	APU completo (%CI, descuento, auxiliares, plantillas)	P-23...P-27	S-24...S-26, S-36	I-06	DONE
011	Presupuesto y versiones	P-28...P-32	S-27...S-32	I-07, I-08	DONE
012	Cronograma y Gantt	P-33...P-36	S-33, S-34	I-08, I-09	DONE
013	Export documentos SERCOP	P-37	S-35	I-10	DONE
014	Panel Super-Admin	P-38...P-42	S-37...S-42	I-11	DONE
015	E2E, accessibility, and CI	—	—	I-01 → I-12	DONE

Coverage check. P-01...P-44 all assigned exactly once · S-01...S-44 all assigned exactly once. The mapping is exhaustive by construction — no process or screen escapes a plan, and no plan claims another's territory.

Execution graph (hard ordering)

The verification gate `pnpm run verify` only exists after 001. Test-suite steps in 005–014 rely on 003. Budget items in 011 reference APUs from 009/010; cronograma activities are auto-imported 1:1 from budget items (011 → 012); export (013) blocks on both. Admin (014) reuses components from 008 and 010 rather than rebuilding them. E2E + CI (015) exercises the finished app.

2 · The architectural decisions that govern the whole codebase

Any plan executor should know these three before writing a line — they are what makes an implementation that *looks* right actually match the thesis's dependent variables.

ADR 9

No calc engine in the client

Every cost, subtotal, HM, CD, CI and CT is computed server-side. RNF-04's "*cálculo en tiempo real*" means **no page reload, cell-commit granularity**: commit a cell → PATCH → repaint from the DTO, with an optimistic *pendiente* state that shows no guessed number.

Rationale: the thesis's headline dependent variable is `exactitud_calculo = 0%` deviation against certified reference cases. A parallel TS implementation would put that at risk for cosmetic gain. Plan 015 mechanises the guard as a CI grep for `toFixed / parseFloat` outside `src/lib/decimal.ts`.

API contract §1

Money travels as decimal strings

Money and percentages are branded `Decimal strings` (`"61.390000"`), never `number` . Percentages travel as fractions (`"0.1800"` = 18 %). Plan 002 brands the type; plan 004 supplies display-only formatters. Never parse to `number` and send it back.

Rationale: JS `number` loses precision at the tail digits SERCOP tolerances depend on.

Codebase design §8

One seam with the backend

`src/api/` is the only place that knows HTTP exists. No `src/features/**` file imports axios or builds a URL. DTOs are hand-transcribed from Apéndice B until the Quarkus backend publishes OpenAPI — at which point `openapi-typescript` replaces the transcription and DTO drift becomes a compile error.

Rationale: keeps HTTP concerns testable at one MSW seam and future-proofs against the still-unbuilt backend.

Design §5.3

Version selector via `?v=` (open decision, closed)

Version was open in `design/02 §5.3` . Resolved to a search param rather than nested routes because `TC-P43-02` already assumed it, and it lets any budget/cronograma/export page share one URL scheme without touching the route table.

Open decisions the plans had to resolve or route around

These were unresolved in `thesis-docs` at plan time. Each plan states its assumption and flags it so the humans' later ruling can be back-propagated.

Open decision	Choice made	Plan
Discount semantics (% vs monto; project + rubro)	Percentage-only; help text mentions both levels exist	007, 010
APU editing model: inline grid vs row-opens-form	Inline (spreadsheet-familiar); SUS pilot may revisit	009
Nested auxiliaries enforced client-side	Not enforced; surface the server's error	010
Central-base propagation (live vs snapshot)	Snapshot; UI shows the price's source	008, 014
Archiving vs deleting a central base	Archivar; UI never says <i>eliminar</i>	014
Gantt: read-only vs drag-to-reschedule	Read-only — v1.1 only asks for “ <i>Gantt simplificado</i> ”	012
Unidades: closed list vs free text	Open catalogue (§17 #10): warn, never block	008
Sync vs async document generation	Sync + stream, isolated in <code>descarga.ts</code>	013
Enforced coverage thresholds	Report only, no gate	003, 015
Design fidelity: low-fi vs Figma	Low-fi; tokens documented as the thesis deliverable	004
Dark mode	shadcn defaults, no investment	004

3 · Traceability to the thesis

Every plan cites the spec repository's stable identifiers so the SPA and the thesis document stay linkable. When a decision is confirmed or changed by the humans, the canonical home is `thesis-docs`; the affected plan is reconciled after.

P-01...P-44	Processes — <code>plan/design/03-procesos-detalle.md</code> . Each plan lists which P-numbers it covers.
S-01...S-44	Screens — <code>plan/design/02-pantallas-flujos.md</code> .
TC / GM	Test cases — <code>plan/quality/02-catalogo-pruebas.md</code> . Every plan lists the TCs its verification step must exercise.
CHK-01...CHK-31	SERCOP conformity items — <code>plan/design/04-export-sercop-spec.md</code> . Measured by plan 013's downloads.
US-nn / I-xx	XP stories and iterations — <code>plan/roadmap/01-plan-iteraciones-xp.md</code> .
D-xx	Process decisions — <code>plan/design/03-procesos-detalle.md</code> §J .
RNF-xx	Non-functional requirements — <code>res/docs/requirements/v1.1-non-functional-requirements.md</code> . RNF-02 (intermodular integrity) is the milestone for plan 012.
ADR 8–10	Architectural decision records — <code>plan/architecture/01-system-architecture.md</code> §9 . ADR 9 (no client calc) is the load-bearing one.

Thesis milestones the plans mechanise directly: **week 10** (plan 009) — reconstruct a real IESS APU in the UI and obtain its exact CT; **week 14** (plan 011) — the IESS tree (33 chapters / 298 rubros) reconstructed; **week 18** (plan 012) — RNF-02 at 100 %; **week 20** (plan 013) — conformity rate CHK-01...CHK-31 measured on the exported files.

4 · Gaps & risks flagged (not papered over)

Six items were surfaced during planning. Each is on the record with the plan that bridges it. None is a blocker for v1.1; each is a candidate for the tutor to flag back to the spec owners.

1. **No `openapi-typescript` source yet.** The *single seam plus DTO-drift-as-compile-error* guarantees depend on OpenAPI, which the still-unbuilt Quarkus backend must publish. Plan 002 bridges with a hand-transcribed Apéndice B and a documented migration path. Until the swap, FE ↔ BE drift is caught by **nothing** except the backend's own API tests.
2. **Firmantes (P-08 / S-11) has no requirements source.** `Firmante` exists in the data model, but v1.1 §2 assigns it neither screen nor process. Plan 007 builds the proposal (a tab in S-09) and isolates it.
3. **No CSV template spec.** P-15 / S-16 step ① says “*descargar la plantilla del tipo*”, but no document defines the columns. Plan 008 derives them from `InsumoCrearRequest` and flags it.
4. **No preview endpoint for exports.** `design/03` P-37 step 3 requires a preview; 07 §8 only offers streaming download. Plan 013 previews PDFs from the downloaded blob and shows a summary card for `xlsx` — at the cost of a second generation round-trip.
5. **Pinned frontend versions were ~1 month stale.** `02-shadcn-components.md` §0.1 was researched June 2026. Plan 001 installs by major line and writes `plans/RESOLVED-VERSIONS.md` for reconciliation.
6. **No per-rubro discount preview endpoint** (unlike the global discount, which has one — 07 §3). Plan 010 shows before/after values rather than fabricating a preview client-side.

Note on the RESOLVED-VERSIONS diff. `pnpm` supersedes `npm` and `oxlint` supersedes `ESLint` relative to the pinned spec. Five packages drifted a patch/minor within the same major line (`shadcn CLI` 4.10 → 4.14, `radix-ui` 1.5 → 1.6.5, `tailwindcss` 4.3.0 → 4.3.3, `TanStack Query` 5.101.0 → 5.101.4, `lucide-react` 1.17 → 1.25). Consistent with plan 001's *install-by-major-line* policy — worth spec-owner sign-off.

Considered and rejected (recorded so they are not re-audited)

- **Client-side document generation** (`ExcelJS` / `react-pdf`). Rejected: ADR 4 / RNF-03 keep SERCOP-format authority server-side.
- **Client-side calc preview for “snappier” totals.** Rejected: ADR 9. The most likely well-intentioned regression, so guarded in CI.
- **Next.js.** Rejected upstream: auth-gated internal app, no SEO need, static Cloudflare Pages deploy.
- **Commercial data-grid / Gantt (AG Grid Enterprise, MUI X Pro, SheetJS Pro).** Rejected on licensing.
- **Decimal arithmetic library** (`decimal.js`, `big.js`). Rejected: the client performs no arithmetic; installing one would only enable what ADR 9 forbids.
- **Per-node query keys for the budget tree.** Rejected: a single cache key lets whole-tree mutation responses repaint write-through totals atomically.
- **Manual activity CRUD in the cronograma.** Rejected: activities are auto-imported 1:1 from budget items — that mechanism is RNF-02's 100 % linkage guarantee.
- **Jest / Enzyme / Cypress.** Rejected upstream (`quality/01` §B1, §B2, §B4).

5 · Meeting talking points

1. **Completeness argument.** P-01...P-44 and S-01...S-44 are each covered exactly once by the fifteen plans (§1). This is the “whole frontend” claim on which the thesis rests.

2. **Load-bearing architectural constraint.** ADR 9 — no client calc engine — is the single decision that keeps `exactitud_calculo = 0 %` defensible; it is enforced by a CI grep, not by convention alone.
3. **Traceability isn't decorative.** Every plan cites P/S/TC/RNF/ADR/US identifiers; the plans and the thesis stay bidirectionally linkable when the spec repo changes.
4. **Honesty about spec debt.** Six unresolved items (§4) are surfaced with the bridging plan named, not hidden behind assumptions. Two of them (openapi-typescript source; firmantes requirements) are appropriate to escalate to the spec owners.
5. **Deferred until the backend exists.** The DTO seam and the export preview both improve materially the day the Quarkus service publishes OpenAPI and an export-preview endpoint — plans 002 and 013 already describe the one-file migration.

Compiled from `plans/README.md`, `plans/RESOLVED-VERSIONS.md`, and the fifteen plan files at repo state 2026-07-24. Every claim in this review is verifiable against the plan file cited in its column.